

ELLWORTH NII ADOTEY ACQUAYE

Senior AI Applications Engineer | Full-Stack Systems | Reliability and Quality

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [Verity](#)

PROFILE

Senior software engineer with 8+ years building and owning production systems across backend, frontend, data, infrastructure and delivery. Currently own Identity, Admin and Payments modules in a live national-utility payments platform: approximately 53,000 lines of hand-written C#, 27 domain entities and 136 API endpoints integrating Apache Fineract, Keycloak and a tier-1 Ghanaian bank.

AI-native, evidence-first engineering practice spanning Claude Code, OpenAI Codex, Anthropic APIs and local Ollama models. Built Verity as an independent TypeScript/Python proof that the same architecture, failure-modeling and quality discipline travels across stacks:

schema-constrained generation, isolated execution, hashed evidence, human approval, staging re-verification, CI and scheduling.

ROLE FIT

AI applications: structured model outputs, prompt/output-schema design, constrained agent workflows, deterministic replay, local/cloud adapters, privacy-aware processing and human-in-the-loop controls.

Full stack: TypeScript (7 yrs), JavaScript (7 yrs), React (6 yrs), Next.js (6 yrs), Node.js, Python/FastAPI, C#/.NET (8 yrs), Angular, Flutter, REST, SSE and OpenAPI.

Data and systems: PostgreSQL, SQL Server, Oracle, Redis Streams/cache/locks, MongoDB, RabbitMQ, Kafka; transactional outbox/inbox, sagas, CQRS, idempotency and event-driven design.

Quality and operations: 540-test production suite, Testcontainers, architecture tests, k6 load/stress/soak gates, Docker, Kubernetes, CI/CD, OpenTelemetry, Jaeger, Prometheus and Grafana.

PROFESSIONAL EXPERIENCE

Senior Software Engineer

Nanti Systems | Accra, Ghana (Remote) | August 2022 - Present

- Own Identity, Admin and Payments modules in a live payments and vending orchestration platform for Ghana's national electricity utility, covering vendor lifecycle, meter/tariff configuration, vending, top-ups, settlement, invoicing, reconciliation, reversals and dispute approvals.
- Designed shared platform infrastructure: CQRS request pipeline, Result-based errors, transactional outbox/inbox with relay jobs, Redis caching, Keycloak/OIDC, API-key issuance, per-vendor rate limiting and YARP gateway boundaries.
- Integrated Apache Fineract for ledger accounts through compensating sagas and a tier-1 Ghanaian bank for collection, transfer, settlement, status reconciliation and authenticated webhook callbacks.
- Authored system architecture and development standards covering clean layering, vertical slices, module boundaries, event contracts, idempotency and recovery; contributed to an 11-service target architecture sized for high national-scale demand.
- Built and maintain 540 unit, integration and architecture tests with real PostgreSQL, RabbitMQ, Redis and Keycloak through Testcontainers; automated boundary rules fail the build when engineering standards are violated.
- Created a fully isolated k6 regression rig with WireMock dependencies and automated thresholds; used it to identify and fix three latent callback, payload-casing and burst-capacity defects before production exposure.
- Mentor engineers, conduct code and architecture reviews, and translate design standards into concrete patterns, anti-patterns and automated enforcement.
- Built CI/CD pipelines with SonarQube and security gates, reducing deployment time by 40% and improving feature-delivery speed by 25%.
- Implemented observability with OpenTelemetry, Jaeger, Prometheus, Grafana, Loki, Seq and structured logging so operations can trace a transaction across participating services.
- Deliver full-stack interfaces in React, Angular and Flutter across web, Android and iOS; design schemas and CQRS read models over PostgreSQL, SQL Server, MongoDB and Redis.
- Designed the platform's Kubernetes migration path, including queue-depth autoscaling that requires deployment changes rather than business-logic rewrites.

Software Developer

Nanti Systems | Accra, Ghana (On-site) | October 2020 - August 2022

- Built enterprise applications across C#/.NET, Angular, React/Next.js, TypeScript, SQL Server, MongoDB and Firebase for utility, telecommunications and healthcare clients.
- Developed third-party API integrations and interface prototypes; promoted to Senior Software Engineer in August 2022.

Computer Programmer

Bista Solutions Ltd. | Accra, Ghana | September 2018 - October 2020

- Contributed to Claims Management and Procurement systems delivered to Ecobank Ghana, West African Ecobank affiliates and GCB Bank, building application, interface and data-access features against banking requirements.
- Delivered procurement and payroll software across C#/.NET, Angular, Flutter, Oracle and SQL Server, including stored procedures, reporting queries and approval workflows.

Earlier experience

Accra, Ghana | 2016 - 2018

- Built an Assets Management System during national service at the Office of the Administrator-General, digitising previously uncaptured state asset records.
- Delivered a Patient Appointment System during a software-development internship at Korle Bu Teaching Hospital.

TECHNICAL TOOLKIT

Languages: C#, TypeScript, JavaScript, SQL, Dart, Java, Python, Go. Frameworks: .NET 10/ASP.NET Core, Next.js, React, Node/Express, FastAPI, Angular, Flutter, Spring Boot, Prisma, EF Core, Dapper.

Architecture: distributed systems, modular monoliths, microservices, DDD, Clean Architecture, vertical slices, CQRS, outbox/inbox, saga compensation, API gateways and event-driven integration.

Security: Keycloak, OAuth2/OIDC, JWT, RBAC/policy controls, HMAC callbacks, API keys, IP allowlisting, rate limiting, secret redaction and OWASP-aligned input boundaries.

SELECTED PROJECTS

Verity - Governed AI Verification Platform | Independent study | 2026

github.com/EllworthAcquaye/verity

- Built an end-to-end TypeScript/Python system that turns versioned intent into schema-constrained declarative checks; model output is replayed for determinism and cannot execute arbitrary code.
- Separated a Next.js/Node control plane from a database-blind FastAPI runner using PostgreSQL transactional outbox, Redis Streams, real HTTP probes and an authenticated result callback.
- Persists redacted canonical evidence with SHA-256 integrity, requires an independent human approval for fixed-file remediation, verifies in isolated staging, and promotes or rolls back automatically.
- Includes Prisma migrations, RBAC personas, shadow nested UI, SSE run updates, reliability scores, database schedules, idempotent CI API, signed evidence export, SHA-pinned GitHub Actions and isolated Docker networks.
- Release proof: 9 architecture tests, 13 Python tests, production build and three live Compose integration gates.

Payments and Vending Orchestration Platform | Module Owner | 2025 - Present

- Live national-utility platform coordinating third-party vendors, bank collection, core-banking ledger, reconciliation, settlement and sales operations. Own Identity, Admin and Payments modules and the secure Flutter admin portal.
- Production stack: C#/.NET 10, Wolverine, RabbitMQ, PostgreSQL, Redis, Keycloak, Apache Fineract, YARP, Quartz, OpenAPI, Docker, Kubernetes and Flutter.

Merchant Settlement Service | Java / Spring Boot | Public source | 2026

[Live API reference](#) | [Source](#)

- Double-entry settlement and reconciliation service with immutable contra-entries, transactional outbox/inbox, Redis idempotency, merchant-scoped authorization and HMAC-signed callbacks.
- Twelve ArchUnit rules, Testcontainers for PostgreSQL/Kafka/Redis, GitHub Actions, Docker Compose and Kubernetes prove the architecture beyond .NET.

EDUCATION AND DEVELOPMENT

Master of Business Administration (MBA), in progress - University of East London, delivered online by Unicaf | August 2026 - 2028

BSc Information Technology - Ghana Communication Technology University, Accra | 2013 - 2017

Java SE - Ghana-India Kofi Annan Centre of Excellence in ICT | 2014

Additional development: Modular Monolith Architecture; ASP.NET Core Clean Architecture and Azure; .NET and Next.js Microservices; Flutter and Dart; customer-centred UX/UI design.

LANGUAGES

English (Native) | German (A1) | French (A1)